

scJDO: Inferring time-varying dynamical operators from single-cell transcriptomic data

David Redd, Sam Green, Tommy W. Terooatea*

Genomics & Bioinformatics Center, Department of Biology, Brigham Young University, Provo, UT, USA

*Correspondence: tommy.terooatea@byu.edu

Abstract

During development, cells traverse gene expression states where their fate becomes transiently sensitive to perturbation — yet existing computational methods provide limited access to when and where this sensitivity peaks along a trajectory. Here we introduce scJDO (single-cell Jacobian Differential Operators), a framework that characterizes how local dynamical sensitivity evolves during cell fate transitions. Crucially, scJDO introduces two independent methodological advances. First, unlike methods that derive Jacobians from splicing-kinetic vector fields, scJDO learns a neural ODE drift field directly from cell state geometry via diffusion score matching, enabling Jacobian analysis on any scRNA-seq dataset regardless of splicing data availability. Second, no existing computational framework treats time-indexed Jacobians as explicit analytical objects for decomposition into recurrent dynamical regimes. scJDO computes local Jacobians — a summary of how small perturbations are amplified or buffered at each point along a trajectory — and decomposes a stack of these operators across pseudotime into a compact set of recurrent dynamical regimes (archetypes) with interpretable temporal activation profiles.

We validate this operator-level framework across five complementary settings: synthetic benchmarks with known ground truth, embedding representation benchmarks, canonical fate bifurcation in hematopoiesis, dense real time-course

reprogramming in induced pluripotent stem cells (iPSCs), and same-cell multiome corroboration in the embryonic brain. Across these settings, scJDO yields reproducible, biologically interpretable sensitivity regimes that reveal the temporal structure of cell fate decisions. Together, scJDO provides an operator-level view of single-cell dynamics that identifies when cell states become locally sensitive to perturbation, including the discovery that failed reprogramming is characterized by a quantitative operator-level signature — absence of the $A1 \rightarrow A2$ handoff — invisible to standard trajectory methods.

Introduction

Cellular identity emerges through dynamic regulatory programs that evolve over developmental trajectories. As cells traverse high-dimensional state spaces, transitions in identity emerge through coupled activity of transcription factors, signaling pathways, and chromatin-associated mechanisms. Single-cell RNA sequencing (scRNA-seq) has transformed our ability to observe these processes, yielding high-resolution snapshots of cellular heterogeneity and developmental progression.

Current computational approaches—including pseudotime inference, RNA velocity, and probabilistic fate mapping—have provided critical insight into trajectory geometry and fate probabilities, but largely operate at the level of where cells are likely to go rather than how local dynamical sensitivity evolves. In particular, they offer limited access to three critical biological questions: where are the perturbation-sensitive regions along a trajectory, which genes or programs drive those regions, and when do distinct regulatory programs activate and hand off? Modern neural SDE and ODE methods learn continuous-time dynamics but have not typically treated time-indexed Jacobians as objects for temporal decomposition.

From a dynamical-systems perspective, differentiation can be viewed as stochastic motion on a high-dimensional manifold, where local properties such as stability, reversibility, and perturbation sensitivity are governed by the Jacobian of the underlying drift field. While local Jacobian analysis is central in dynamical systems, existing single-cell implementations (e.g., Dynamo, SpliceJAC) derive Jacobians from splicing-kinetic vector fields, which require metabolic labeling or explicit assumptions about RNA production and degradation. Furthermore, these methods treat Jacobians as per-cell regulatory snapshots. No existing computational framework treats time-

indexed Jacobians as explicit analytical objects for identifying recurrent operator regimes across developmental progression.

Here we introduce scJDO (single-cell Jacobian Differential Operators), a framework that occupies this uncontested niche. Unlike methods that derive Jacobians from splicing-kinetic vector fields, scJDO learns a neural ODE drift field directly from cell state geometry via diffusion score matching, enabling Jacobian analysis on any scRNA-seq dataset regardless of splicing data availability. Dynamo computes local Jacobians at individual cell states to identify regulatory interactions; scJDO instead treats the temporal sequence of Jacobians as a primary analytical object, decomposing the resulting tensor into recurrent operator archetypes with interpretable activation profiles across developmental progression. By stacking these operators across pseudotime and decomposing the resulting tensor, scJDO identifies recurrent operator archetypes—reusable local dynamical regimes—and estimates when they become active, overlap, or recede.

This operator-centric view provides a new lens on single-cell data. We test the utility of this framework across five distinct validation settings: synthetic benchmarks, embedding representation benchmarking, branching hematopoiesis, real time-course progression, and same-cell chromatin corroboration. In this way, scJDO provides a framework for studying when cell states appear dynamically buffered, plastic, or sensitive to perturbation.

Results

scJDO infers time-resolved operator representations of local sensitivity

The scJDO framework transforms static single-cell measurements into an operator-centric representation of cellular dynamics through four stages (Fig. 1a). scJDO’s operator inference is performed independently and does not require predefined basins.

Stage 1: Drift field estimation. We model cellular state evolution in a latent space as a stochastic differential equation:

$$dx(t) = f(x(t), t) dt + \sigma dW_t$$

where $x(t) \in \mathbb{R}^d$ denotes the latent cellular state, $f(\cdot, t)$ is a learned drift field, and W_t is a Wiener process. The drift is parameterized using a hybrid architecture that combines diffusion score matching with residual deterministic components and optional RNA velocity guidance, yielding a smooth vector field suitable for operator analysis (Methods). When used, RNA velocity acts as a weak directional prior on f and does not define macrostates, terminal basins, or archetypes.

Stage 2: Local operator inference. We compute local Jacobian operators along inferred progression by automatic differentiation:

$$J(x, t) = \frac{\partial f(x, t)}{\partial x} \in \mathbb{R}^{d \times d}$$

These operators provide a **comparative summary of local linearized sensitivity**: how small perturbations are predicted to be amplified or buffered around a given point in state space (Fig. 1b). These modes arise from derivatives of the learned drift field, reflecting inferred local response properties rather than static covariance structure.

Stage 3: Temporal tensor construction. We stack Jacobians across pseudotime windows to form a temporal Jacobian tensor:

$$\mathcal{J} \in \mathbb{R}^{T \times d \times d}$$

where T indexes pseudotime windows and d is the latent dimension (Fig. 1c). This tensor summarizes how local dynamical sensitivity evolves during differentiation.

Stage 4: Archetype decomposition. To obtain a compact and interpretable representation, we decompose $\mathcal{J}$ via constrained low-rank factorization:

$$J(t) \approx \sum_{k=1}^K a_k(t) A_k$$

where $A_k \in \mathbb{R}^{d \times d}$ are operator archetypes—recurrent Jacobian regimes—and $a_k(t)$ are their time-varying activation profiles (Fig. 1d). This decomposition reveals both the repertoire of recurrent local dynamical regimes and the temporal patterns by which they are coordinated.

Box 1 | What scJDO identifies

scJDO robustly identifies (empirically invariant across seeds and architectures):

- Relative operator structure (which genes co-vary in their sensitivity)
- Temporal ordering of operator regimes (e.g., sequential handoffs)
- Timing of local-sensitivity peaks relative to developmental progression
- Top-ranked genes associated with locally sensitive directions (median Jaccard ≥ 0.81)

scJDO is not designed to identify:

- Absolute biological Jacobians (exact entries of the regulatory matrix)
- Absolute eigenvalues (scale depends on pseudotime parameterization)
- Causal regulatory interaction strengths

Rather than recovering a unique underlying regulatory Jacobian, scJDO targets features of local dynamical structure that are reproducible across independently trained models and representations.

Throughout this paper, “instability” refers strictly to the **local mathematical sensitivity** of the inferred transcriptomic drift field—specifically, the sign and magnitude of the dominant real eigenvalue of the local Jacobian—and not to physical instability or direct measurement of regulatory mechanism.

Synthetic benchmarks validate core mathematical claims

Before interpreting biological datasets, we validated the core mathematical claims of scJDO on synthetic simulations where the true dynamics are known analytically. This ensures that the framework recovers expected operator structures without confounding biological complexity.

We tested scJDO across four synthetic scenarios:

1. **Drift recovery on a double-well potential:** scJDO accurately recovered the drift field, with an endpoint mean squared error (MSE) of 0.0005, and 100% of simulated trajectories reaching the correct attractor.
2. **Jacobian recovery:** The inferred local Jacobians strongly correlated with the true analytical Jacobians (eigenvalue correlation = 0.758), successfully recovering the sign transition at the saddle point. However, instability-gene recovery was

less precise, highlighting that while the operator structure is robust, projecting back to individual genes requires caution.

3. **Archetype recovery:** On a system with known stable, bifurcating, and oscillatory regimes, scJDO's archetype decomposition successfully recovered three distinct regimes with a low reconstruction error (0.228).
4. **Schrödinger bridge transport:** When transporting between two known Gaussian marginals, scJDO's Schrödinger bridge formulation accurately recovered the optimal transport plan, achieving a Wasserstein distance (W_2) of 0.065 to the target, closely matching the analytic truth.

These synthetic benchmarks confirm that scJDO accurately infers drift fields, local Jacobians, and temporal archetypes in systems with known ground truth.

Factor Analysis provides a robust, interpretable latent space for operator inference

To determine the optimal latent representation for scJDO, we benchmarked eight embedding methods (PCA, Factor Analysis, ICA, TruncatedSVD, DiffMap, LDVAE, scVI, and PLS) on a bone marrow hematopoiesis dataset. We evaluated each method based on marker enrichment, branch specificity, stability across random seeds, timing spread, and drift field goodness-of-fit (R^2).

While supervised Partial Least Squares (PLS) achieved the highest composite score (0.670), Factor Analysis (FA) emerged as the strongest unsupervised method (0.656), outperforming nonlinear methods like scVI (0.597) and LDVAE (0.586), as well as the PCA baseline (0.553). FA provided highly specific branch instability signatures (specificity = 0.911, mean Jaccard = 0.089) and successfully recovered canonical markers in the top-30 instability genes for the Erythroid (50%) and Monocyte (71%) branches. Consequently, we adopted FA as the primary latent representation for subsequent scJDO analyses.

Hematopoiesis reveals transient local sensitivity preceding fate bifurcation

We applied scJDO to the Paul15 hematopoiesis dataset, which spans differentiation from progenitors to erythroid, dendritic cell (DC), and monocyte lineages. Using the FA latent space and Palantir pseudotime, we inferred branch-separated drift fields that accurately captured the lineage progression ($R^2 > 0.99$).

Jacobian analysis revealed transient emergence of locally sensitive modes preceding lineage bifurcations. For the Erythroid and DC branches, peak sensitivity occurred late in pseudotime (pt = 0.707 and 0.822, respectively), corresponding to the commitment phase. The top instability genes for the Erythroid branch included canonical markers such as *HBB*, *AHSP*, and *CA1*, with *GATA1* identified as a top transcription factor regulator. For the DC branch, the instability signature was dominated by cell-cycle and proliferation genes (*MKI67*, *TOP2A*, *BIRC5*), consistent with the proliferative burst known to occur during common dendritic cell progenitor (CDP) commitment, alongside the canonical DC regulator *IRF8*.

These results illustrate how scJDO can summarize lineage progression in terms of evolving local dynamical regimes, capturing the timing and molecular drivers of lineage commitment.

Dense reprogramming time-course analysis reveals branch-specific operator handoffs during successful versus diverted reprogramming

To address whether scJDO's operator regimes are robust to real time-course sampling rather than only snapshot-derived pseudotime, we applied the framework to a densely sampled induced pluripotent stem cell (iPSC) reprogramming dataset (SCP295). This dataset captures the continuous transition from mouse embryonic fibroblasts (MEFs) to iPSCs over 18 days, with known branch points separating successful reprogramming from alternative fates such as stromal diversion.

The inferred drift field, when mapped onto the force-directed layout embedding colored by actual experimental time, accurately captured the temporal progression of reprogramming (Fig. 4a). We identified four major cell fate branches: Early MEF, MET corridor, Stromal diversion, and successful iPSC (Fig. 4b). The critical divergence point between the successful MET corridor and the Stromal diversion occurs around day 8.8, where the branches show statistically detectable divergence in mean local sensitivity.

Local operator sensitivity analysis revealed a transient window of elevated sensitivity (positive eigenvalues) corresponding to the commitment phase of early activation, specifically around the MET onset and the divergence point (Fig. 4c). Archetype activation profiles demonstrated a sequential handoff from an early MEF exit program (Archetype 1) to a late iPSC program (Archetype 2) along the successful trajectory (Fig. 4d). Crucially, the operator handoffs differed significantly between the successful MET corridor and the stromal diversion branch. The A1→A2 handoff crossover occurs at pseudotime 0.691 in the MET corridor and 0.696 in the iPSC branch, but is **absent in**

the Stromal diversion branch (no crossover detected across the full pseudotime range). Crucially, this handoff failure is detectable at pseudotime 0.6, well before the branches become visibly separated in the force-directed layout (day 8.8), confirming that failure to execute the A1→A2 handoff is a quantitative, predictive feature of the diverted fate rather than a post-hoc consequence of branch assignment.

We evaluated the overlap between the top 50 driving genes of each scJDO archetype and established biological gene modules from the literature. Archetype 1 strongly aligned with the MEF identity module, Archetype 2 with the Pluripotency and Metabolic shift modules, and Archetype 4 with the Metabolic shift module. This confirms that the unsupervised mathematical operators capture distinct, known biological programs and that scJDO can resolve time-resolved success versus diversion in a real time-course setting.

Same-cell multiome analysis provides orthogonal chromatin support for local-sensitivity peaks and archetype-associated programs

To provide orthogonal validation for these transcriptomic sensitivity regimes, we analyzed a paired multiome dataset (10x Genomics E18 Mouse Brain Fresh 5k Multiome) containing both scRNA-seq and scATAC-seq measurements from the same cells. We focused on the neurogenic lineage, comprising radial glia (progenitors), intermediate progenitor cells (IPCs), and excitatory neurons.

Local sensitivity computed from the RNA-derived drift field revealed distinct peaks concentrated in the narrow pseudotime window corresponding to the radial glia–IPC transition. Archetype activation profiles revealed a handoff from progenitor-enriched programs at early pseudotime toward neuronal programs at later pseudotime, with the sensitivity peak coinciding with the transition window.

Crucially, we found strong same-cell chromatin support for these RNA-derived archetypes. The radial glia/progenitor archetype showed the strongest positive correlation with mean promoter ATAC accessibility across the same cells ($r = 0.85$, $p < 10^{-22}$), consistent with open chromatin at progenitor-associated loci early in the trajectory. Conversely, the early neuronal archetype showed a negative ATAC correlation ($r = -0.52$, $p < 10^{-6}$), reflecting progressive chromatin remodeling as cells exit the progenitor state. Furthermore, archetype activation curves were strongly correlated with specific TF motif scores from the ATAC modality (e.g., Atoh1, OLIG1, BHLHE22), providing direct chromatin-level support for the RNA-derived archetypes.

Same-cell chromatin corroboration provides orthogonal, modality-independent support for RNA-derived operator archetypes.

Discussion

scJDO establishes a new paradigm for analyzing single-cell dynamics by treating time-varying Jacobians as explicit analytical objects. This operator-centric framework reveals biological insights that are invisible to standard trajectory and fate-mapping tools. The central contribution of scJDO is to represent single-cell dynamics at the level of time-resolved local sensitivity, enabling decomposition into a small set of reusable operator regimes with reproducible temporal structure.

The power of this approach is most clearly demonstrated in the iPSC reprogramming time-course. While standard methods can identify the divergence between successful reprogramming and stromal diversion, scJDO reveals the underlying dynamical failure: the successful trajectory executes a clean sequential handoff from an early MEF exit program (Archetype 1) to a late iPSC program (Archetype 2), whereas the stromal diversion branch completely fails to execute this handoff. This provides a quantitative, operator-level signature of developmental failure that goes beyond differential expression.

Similarly, in hematopoiesis, scJDO identifies transient peaks in local sensitivity that precede fate bifurcations and program handoffs. These peaks are not merely mathematical artifacts; they are strongly enriched for known lineage regulators and, as shown in the embryonic brain multiome data, align with elevated chromatin accessibility in transitional states. By organizing local sensitivity into temporally structured, biologically interpretable regimes, scJDO provides a principled way to generate candidate perturbation hypotheses.

Relation to existing dynamical approaches

scJDO occupies an uncontested niche in the single-cell analysis ecosystem. While transition-operator frameworks (e.g., CellRank, Palantir) identify macrostates and fate probabilities, and continuous-time models (e.g., VeloVAE, neural ODEs) learn trajectory geometry, no existing method targets the local dynamical sensitivity encoded in time-indexed Jacobians. scJDO is designed to complement these approaches (Table 1).

Table 1. Decision guide for single-cell dynamics methods.

Primary question	Recommended method	scJDO role
Where will this cell end up? What are the fate probabilities?	CellRank, Palantir, Slingshot	Input: use CellRank macrostates to annotate archetype activation regions
How fast are individual genes changing? What are RNA velocity estimates?	scVelo, Dynamo, UniTVelo	Parallel: scJDO uses velocity as an optional weak prior; run independently
What is the continuous-time dynamics model?	VeloVAE, neural ODE	Parallel: scJDO can be run on the same latent space; outputs are complementary
Which pseudotime regions are most sensitive to perturbation?	scJDO	Primary use case
Which genes drive sensitivity at transition points?	scJDO locally sensitive modes	Primary use case
When do distinct regulatory programs activate and hand off?	scJDO archetype decomposition	Primary use case
What is the chromatin accessibility landscape?	ArchR, Signac	Orthogonal validation: compare scATAC peaks to local-sensitivity peaks

scJDO operates downstream of trajectory inference: it requires a pseudotime ordering as input and returns an operator-level summary of how local dynamical sensitivity evolves along that ordering. Users should first establish a biologically meaningful trajectory using their preferred method, then apply scJDO to characterize the sensitivity structure of that trajectory.

Failure modes

Intellectual honesty about the boundaries of a method is as important as demonstrating its strengths. We identify five conditions under which scJDO conclusions should be treated with particular caution or avoided entirely.

Static perturbation screens. scJDO is not appropriate for static perturbation screens lacking a biologically meaningful developmental or response trajectory; in such settings, pseudotime is arbitrary and the inferred Jacobian operators can become degenerate. The method assumes a structured progression where local sensitivity has biological meaning.

Sparse or narrow pseudotime coverage. scJDO requires sufficient cell density across the pseudotime axis to estimate a smooth drift field. In datasets where cells are concentrated at a few discrete time points or where large pseudotime intervals are sparsely sampled, the learned drift field will be poorly constrained in those regions, and the resulting Jacobians will reflect model extrapolation rather than data.

Cyclic or disconnected manifolds. The current implementation assumes a roughly tree-structured or linear progression. Cyclic trajectories (e.g., cell-cycle dynamics) and disconnected manifolds (e.g., distinct tissue compartments analyzed jointly) violate this assumption. Applying scJDO to such data may produce archetypes that reflect the topology of the manifold rather than genuine dynamical regimes.

Strong nonlinear coordinate distortions. Archetype identities are empirically robust to linear rotations and changes in latent dimension, and to architectural variation as described above, but they are not guaranteed to be robust to strong nonlinear reparameterizations of latent space.

Systems dominated by cell-cell interactions. scJDO models cell-autonomous dynamics: the drift field $f(x, t)$ depends only on the state of the individual cell and pseudotime, not on the states of neighboring cells. In systems where cell fate is primarily governed by juxtacrine or paracrine signaling, microenvironmental gradients, or spatial coupling, the cell-autonomous model will be misspecified, and the inferred operators will not capture the dominant regulatory logic.

Limitations

scJDO infers local linearized sensitivity of a learned drift field in latent space; it is not a direct measurement of physical regulatory interactions and should not be interpreted as a causal gene regulatory network. All biological language in this manuscript is intentionally cautious: enrichment results are supportive rather than confirmatory, and operator structure is model-conditional rather than a ground-truth property of the underlying biology. Biological interpretation therefore depends on projection to gene space and on external validation.

Because scJDO analyzes Jacobians of a learned drift field, its conclusions depend on how well that field is constrained by the available data and modeling assumptions. Temporal indexing and model choice remain important sources of uncertainty. The current implementation primarily assumes cell-autonomous dynamics and does not explicitly model cell–cell communication. In addition, inferred operators should be interpreted as transcriptomic dynamical surrogates rather than direct measurements of underlying regulatory mechanism. Finally, while locally sensitive modes identify candidate perturbation-sensitive directions, causal conclusions will require targeted perturbations and prospective validation.

Overall, scJDO illustrates how operator-based analyses can enrich interpretation of single-cell data by shifting attention from trajectory geometry alone to evolving local sensitivity. We anticipate that this perspective will be useful for building dynamical complements to expression atlases and for generating experimentally testable hypotheses about when and where cell fate decisions become locally sensitive to perturbation.

Methods

Data preprocessing and latent space embedding

Raw gene expression matrices were normalized to counts per 10,000, log-transformed, and restricted to the top 2,000 highly variable genes (Scanpy v1.9.3; Seurat flavor). Dimensionality reduction used Factor Analysis (FA) with 30 components, which was selected based on comprehensive benchmarking against PCA, ICA, TruncatedSVD, DiffMap, LDVAE, scVI, and PLS. FA provided the most biologically interpretable branch instability signatures and robust marker recovery among unsupervised methods.

Temporal coordinate inference and trajectory sampling

Diffusion pseudotime (DPT) was computed on a k -nearest-neighbor graph ($k = 15$; Euclidean distance in FA space). Robustness comparisons used Palantir (v1.0.1) and Slingshot (v2.4.0). Pseudotime was discretized into 100 overlapping windows (80% overlap), and representative latent states were obtained by averaging cells within each window. For temporal alignment across methods and runs, activation profiles were aligned using dynamic time warping (DTW).

Relation to velocity-based fate mapping

Velocity- and transition-operator methods infer macrostates, terminal basins, and fate probabilities by constructing transition operators informed by RNA velocity or related directional information. scJDO instead targets local operator structure by learning a smooth drift field and computing time-indexed Jacobians along inferred progression. Unless otherwise stated, scJDO results are reported without using CellRank-derived macrostates for archetype definition; macrostate annotations may be overlaid for visualization only. Velocity guidance, when used, is treated as a weak directional prior.

Drift field parameterization and learning

Cellular dynamics were modeled as:

$$dx(t) = f(x(t), t) dt + \sigma dW_t$$

The drift f was parameterized by a time-conditioned multilayer perceptron with four layers (width 256; SiLU activations), residual skip connections, FiLM-style time conditioning, and spectral normalization on the final layer. The model was trained with diffusion score matching plus optional velocity guidance using AdamW (learning rate 2×10^{-4} , weight decay 10^{-4}), batch size 1,024, cosine decay schedule, for 50,000 iterations on an NVIDIA A100 GPU.

Jacobian inference and eigenanalysis

Local Jacobians were computed via automatic differentiation in PyTorch (v2.1.0):

$$J(x, t) = \nabla_x f(x, t)$$

We performed eigen-decomposition using `torch.linalg.eig` without symmetrization. Modes were classified as stable, plastic, or locally sensitive based on the maximum real eigenvalue: stable if $\text{Re}(\lambda_{\max}) \leq -0.05$, locally sensitive if $\text{Re}(\lambda_{\max}) \geq +0.05$, and plastic otherwise. Eigenvectors were projected to gene space using FA loadings.

Temporal Jacobian tensor construction and archetype decomposition

Jacobians across pseudotime were stacked into $\mathcal{J} \in \mathbb{R}^{T \times d \times d}$. Low-rank decomposition used constrained alternating least-squares factorization with non-negativity on

activations and K selected via a consensus procedure. Optimization used L-BFGS with 20 restarts, selecting the solution with highest reconstruction R^2 .

Schrödinger Bridge formulation

For the K562 CRISPRi Perturb-seq dataset, we employed a Schrödinger Bridge (SB) formulation to regularize the drift field. The SB objective minimizes the Kullback-Leibler divergence to a reference Wiener process, subject to matching the empirical marginal distributions at the initial (non-targeting control) and final (perturbed) time points. This was implemented using an alternating Sinkhorn and score-matching algorithm, anchoring the inferred trajectory to the observed biological endpoints.

SCP295 iPSC Reprogramming Analysis

We analyzed 27,538 cells from the SCP295 dataset (Schiebinger et al. 2019) from serum and Dox reprogramming conditions. Four frozen branch assignments were derived from published cell-set annotations: Early MEF, MET corridor, Stromal diversion, and successful iPSC. The primary root was defined at day 0. Drift fields were fit independently for each branch using the FA latent space.

Multitome analysis

For the 10x Genomics E18 Mouse Brain Fresh 5k Multitome dataset, RNA and ATAC modalities were processed jointly. The dataset comprised 4,628 cells after QC. The neurogenic lineage (3,725 cells) was identified by co-expression of progenitor and neuronal markers. A joint RNA+ATAC latent embedding was computed from the concatenation of RNA PCA and ATAC LSI. Diffusion pseudotime (DPT) was computed on the shared RNA+ATAC nearest-neighbor graph. The drift field and local Jacobians were inferred exclusively from the RNA modality. Archetype activation profiles were correlated with mean promoter accessibility and TF motif scores (computed via Signac, JASPAR 2020) across the same cells using Spearman rank correlation.

Author Contributions

Using the CRediT (Contributor Roles Taxonomy) framework:

David Redd: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

Sam Green: Methodology, Software, Validation, Data curation, Visualization, Writing – review & editing.

Tommy W. Terooatea: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Competing Interests

The authors declare no competing interests.

Data Availability

The Paul15 hematopoiesis dataset is available via the Scanpy package. The SCP295 iPSC reprogramming dataset is available via the Single Cell Portal. The 10x Genomics E18 Mouse Brain Fresh 5k Multiome dataset is available at 10xgenomics.com. The K562 CRISPRi Perturb-seq dataset is available from 10x Genomics. All accession numbers and data sources are listed in Supplementary Table 1.

Code Availability

The scJDO package is available at <https://github.com/manarai/scJDO>. Figure-generating notebooks are provided in the `Figures_notebook/` directory.